🧩 What is a Module in Python?

A **module** in Python is simply a **file containing Python code** — functions, variables, classes, etc. It allows you to **organize code** logically and **reuse it** across multiple programs.

👉 Any file with a .py extension is considered a **Python module**.

🧠 Why Use Modules?

- ✅ Makes your code organized and modular
- ✅ Reuse functions and classes
- ✅ Avoid code duplication
- ✅ Easy maintenance and debugging

📘 Example 1: Creating and Using a Custom Module

Let's say you have a file named **mymath.py**:

```
# mymath.py

def add(a, b):
    return a + b


def subtract(a, b):
    return a - b
```

Now you can use this module in another file **main.py**:

```
# main.py

import mymath


print(mymath.add(10, 5))       # Output: 15

print(mymath.subtract(10, 5))  # Output: 5
```

🧠 **Note:** Both files must be in the same folder, or you must specify the correct path.

⚙️ **Types of Modules**

1. **Built-in Modules** (come with Python)
   Example:
2. import math
3. import random
4. import datetime
5.
6. print(math.sqrt(16)) # 4.0
7. print(random.randint(1, 10)) # Random number between 1–10
8. print(datetime.date.today()) # Current date
9. **User-defined Modules**
   (Like the mymath.py example above)
10. **External Modules (3rd Party)**
    Installed via **pip**.
    Example:
11. pip install requests

Then use it:

```
import requests

response = requests.get("https://api.github.com")

print(response.status_code)
```

🧩 **Import Variations**

You can import modules in several ways:

1️⃣ **Import the whole module**

```
import math

print(math.pi)
```

2️⃣ Import specific functions or variables

```
from math import sqrt, pi

print(sqrt(9))

print(pi)
```

3️⃣ Import with alias

```
import math as m

print(m.pow(2, 3))
```

4️⃣ Import everything (⚠ not recommended)

```
from math import *

print(sin(0))
```

📦 Checking Installed Modules

List installed packages:

```
pip list
```

Check if a module exists:

```
help('modules')
```

🧰 Useful Built-in Modules

Module	Purpose
math	Mathematical functions
datetime	Dates and times
os	Interact with operating system
sys	Access system-specific parameters
random	Generate random numbers

Module	Purpose
json	Work with JSON data
re	Regular expressions
csv	Handle CSV files
collections	Specialized data structures
statistics	Statistical calculations

👷 **Example: Using Multiple Modules**

```
import math
import random
import datetime

num = random.randint(1, 100)
print(f"Square root of {num} is {math.sqrt(num)}")
print("Today's date:", datetime.date.today())
```